

Flume mooring specification — OSU Large Wave Flume (HWRL), 1:50 test articles

REV C, 2026-09-25: two cord sets on the same anchors and attachment points (supersedes rev B

fb6ba5b, whose design is the operational set). Generated by build_spec.py and

mooring_figures.py from FloatSim results; basis DESIGN-BASIS.md Phases F–G.

Summary

	single buoy	cluster	4×4 platform
anchors (under water, on the side walls)	x ±5.0, y ±1.83, z -0.50 m	x ±5.0, y ±1.83, z -0.15 m	x ±5.0, y ±1.83, z -0.15 m
attachment points	slender radial collar r = 0.2 m on the spar, 4 points at z = -0.50 m	each of the 4 spars at z = -0.15 m	the 8 up-/down-stream row spars at z = -0.15 m (4 two-leg bridles)
lines (legs)	4	4	8
pretension at rest per line/leg, both sets (set by tension or calm tilt)	2.40 N	1.44 N	1.42 N
operational cord (response tests, H ≤ 0.12 m)	k 2.01 N/m (−13 / +30 %), L0 3.930 m (nominal T0 2.40 N), pre-stretch at rest 1195 mm, elongation ≥ 2.21 m (56 % of L0)	k 4.15 N/m (−13 / +30 %), L0 4.589 m (nominal T0 1.49 N), pre-stretch at rest 346 mm, elongation ≥ 0.63 m (14 % of L0)	k 8.32 N/m (−13 / +30 %), L0 4.038 / 4.208 m (nominal T0 1.54 N), pre-stretch at rest 171 mm, elongation ≥ 0.42 m (10 % of L0)
extreme cord (H = 0.2–0.5 m)	the same cord (one set)	k ×5 : k 20.75 N/m (minimum; −0 / +30 %), L0 4.866 m (nominal T0 1.73 N), pre-stretch at rest 69 mm, elongation ≥ 1.33 m (27 % of L0)	k ×6 : k 49.91 N/m (minimum; −0 / +30 %), L0 4.181 / 4.350 m (nominal T0 2.14 N), pre-stretch at rest 28 mm, elongation ≥ 1.13 m (27 % of L0)
max line tension: operational / extreme	3.54 N	2.09 / 22.12 N	2.81 / 45.27 N
anchor working load ×3 (max of both sets)	10.6 N	66.4 N	267.4 N
calm static tilt, FloatSim: operational / extreme	0.00° (collar)	1.00° / 1.00°	1.00° / 1.00°
H = 0.5 m max surge (limit +1.0 m)	0.68 m (predicted cases)	0.97 m (extreme set; operational set 3.39 m)	0.95 m (extreme set; operational set 3.53 m)
tilt-period shift vs unmoored: operational (limit ζ/3) / extreme (declared)	-1.37 % (≤ 1.42 %)	-0.73 % (≤ 1.19 %) / -3.77 %	-0.72 % (≤ 1.05 %) / -4.65 %

- **Two cord sets, one rig.** Swap the cords between the response series (operational set, H ≤ 0.12 m) and the extreme series (extreme set, H = 0.2–0.5 m). The anchors, the attachment points and the AT-REST tension stay the same, so the calm geometry is the same.
- **Installation checks, for each set** (§5):
 - the calm static tilt, which checks the pretension (1° at the design at-rest tension on the cluster and platform);
 - a static pull against the predicted pull stiffness;
 - surge and sway decays against the predicted periods.
- **Extremes:**
 - Run them in increasing H with live offset monitoring.
 - Run the buoy's extremes LAST, with a height ramp, and watch its yaw: 3 of its cases have no reliable FloatSim prediction.
- **Buoy collar:** a SLENDER rod or wire spreader, round section ≤ 25 mm. A disk would add ≈ 21 kg.
- **Facility assumptions F1–F7** (next page) must be confirmed with HWRL before any hardware is bought. F1: underwater wall anchors.

Facility assumptions to CONFIRM WITH HWRL BEFORE BUYING HARDWARE

None of these is a confirmed facility fact. The design rests on all of them.

#	Assumption	Value this specification uses	If it proves false
F1	Anchors can be fixed on the side walls under water, at each article's attachment depth	anchors at $x = \pm 5.0$ m, $y = \pm 1.83$ m, $z = -0.50$ m (buoy), -0.15 m (cluster), -0.15 m (platform)	A higher or lower anchor slopes the line. Its vertical pull at the attachment adds calm tilt and tilt coupling, so the attachment sweep must be re-run for the real anchor depth.
F2	Wall anchor rating	per-anchor working-load limit, the larger of the two cord sets ($\times 3$ on the FloatSim peak): buoy 10.6 N, cluster 66.4 N, platform 267.4 N	Lower than these is unlikely to bind; confirm.
F3	Wave height per period, steepness	$H/\lambda \leq 0.08$ at 2.7 m depth: H 0.5 m only for $T \geq 2.01$ s (and from 2.35 s by the 3° tilt rule), H 0.35 m for $T \geq 1.675$ s	Cases outside the wavemaker envelope drop out.
F4	Water depth	2.7 m	The drift bound falls for deeper water; periods unchanged.
F5	Test-section position vs the wavemaker and beach	a clean regular-wave window $\geq 2\text{--}3$ min at the article, which sits on the flume centreline	Shorter windows do not settle the lightly damped resonances ($\zeta \approx 3\text{--}4\%$).
F6	Tracking field of view	surge $-0.3 \dots +1.0$ m (down-flume), sway ± 0.1 m, plus the article footprint (platform 2.06 m wide at 45°)	With the extreme set, H = 0.5 m stays within +1.0 m at the two worst periods run (§3b). A smaller window needs a stiffer extreme set.
F7	Soft-mooring cord, usable submerged, in two stiffnesses per article	elastic cord: the operational set (secant stiffness within $-13 / +30\%$ of target) and the extreme set (a MINIMUM stiffness, $-0 / +30\%$), stiffness and elongation per the tables; submerged weight ≈ 0.02 N/m	Without it, the lines need farther anchors (longer lines, lower strain) or another soft element. Redesign.

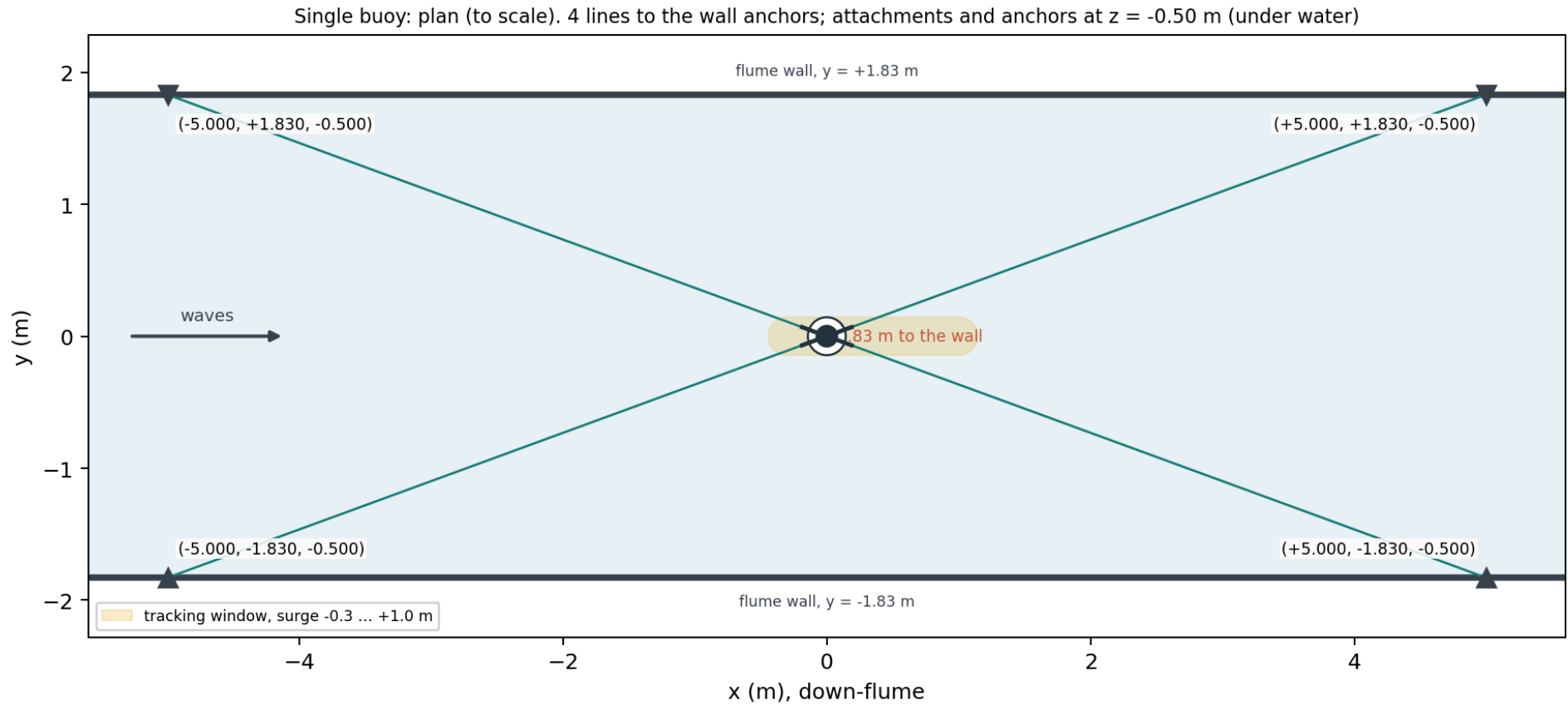
1. Layout and the two cord sets

- An X-spread of four soft lines to the side walls, 20.1° off the flume axis. The platform's four lines are 2-leg bridles (8 legs) to the two row spars of each up- and down-stream half-row.
- **Attachments are below still water, and each anchor is at the same depth**, so every line runs level (figures in §2):
- **single buoy**: a slender radial collar of $r = 0.2$ m on the spar at $z = -0.50$ m, one line at the point facing each anchor (§4);
- **cluster**: each spar at $z = -0.15$ m;
- **platform**: the 8 up- and down-stream row spars at $z = -0.15$ m.
- Each line or leg is an **elastic cord**, with short stiff leads or shackles if needed, of unstretched length L_0 . It is pre-stretched to the pretension T_0 when attached: the tension at rest $\approx k \times (\text{chord} - L_0)$.
- **Two cord sets** (rev C):
- The **operational set** is rev B unchanged. It is used for every response test ($H \leq 0.12$ m), and it disturbs the article least (§3a). Calm static tilt $\leq 1^\circ$ is now CONFIRMED by Xabier.
- The **extreme set** is a stiffer cord at the same at-rest tension on the same anchors and attachment points. It is used for the $H = 0.2\text{--}0.5$ m load and survival tests, and keeps the article in the tracking window (§3b).
- The buoy needs no second set: its predicted $H = 0.5$ m offsets stay inside $+1.0$ m.
- Why two sets:
- Rev B's decision rule omitted the confirmed excursion criterion (Phase B criterion 4).
- The conflict itself is real: minimal interference needs soft lines, and station-keeping in extreme waves needs stiff ones.
- Two cord sets resolve it without compromising either design.
- Frame for all coordinates (the TRUE flume frame): the origin is at the article's calm centre, on the flume centreline, at still water. x runs down-flume (the wave direction), y across, z up.
- What is conservative in these numbers:
- the **drift sum**: the full recorded fixed-body drift bound, applied in-run at each spar's waterline, plus FloatSim's own mean wave force on the moving body;
- **working loads $\times 3$** and **elongation capacity $\times 1.25$** on the worst predicted FloatSim values.

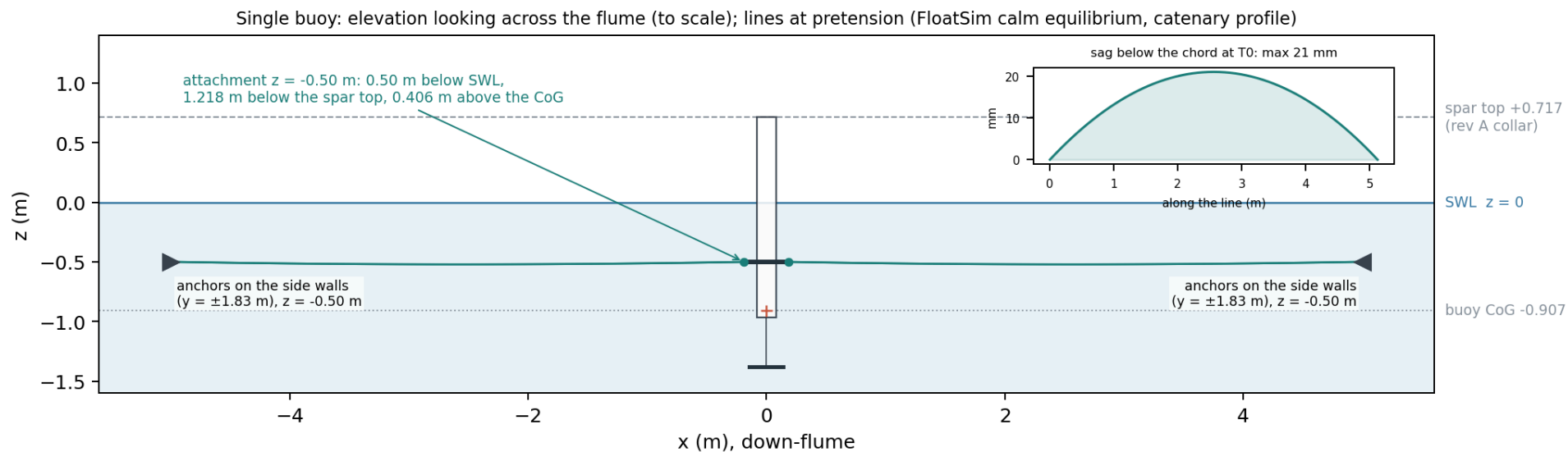
2. Per-article arrangement and line tables

The **up-flume lines (anchors at $x = -5$ m)** carry the drift and set every peak value. **Buy and rig every line of a set to its up-flume values**, so the lines are interchangeable.

2a. Single buoy



Single buoy: plan, to scale (mooring_figures.py, from spec_statics.json)



Single buoy: elevation looking across the flume, to scale

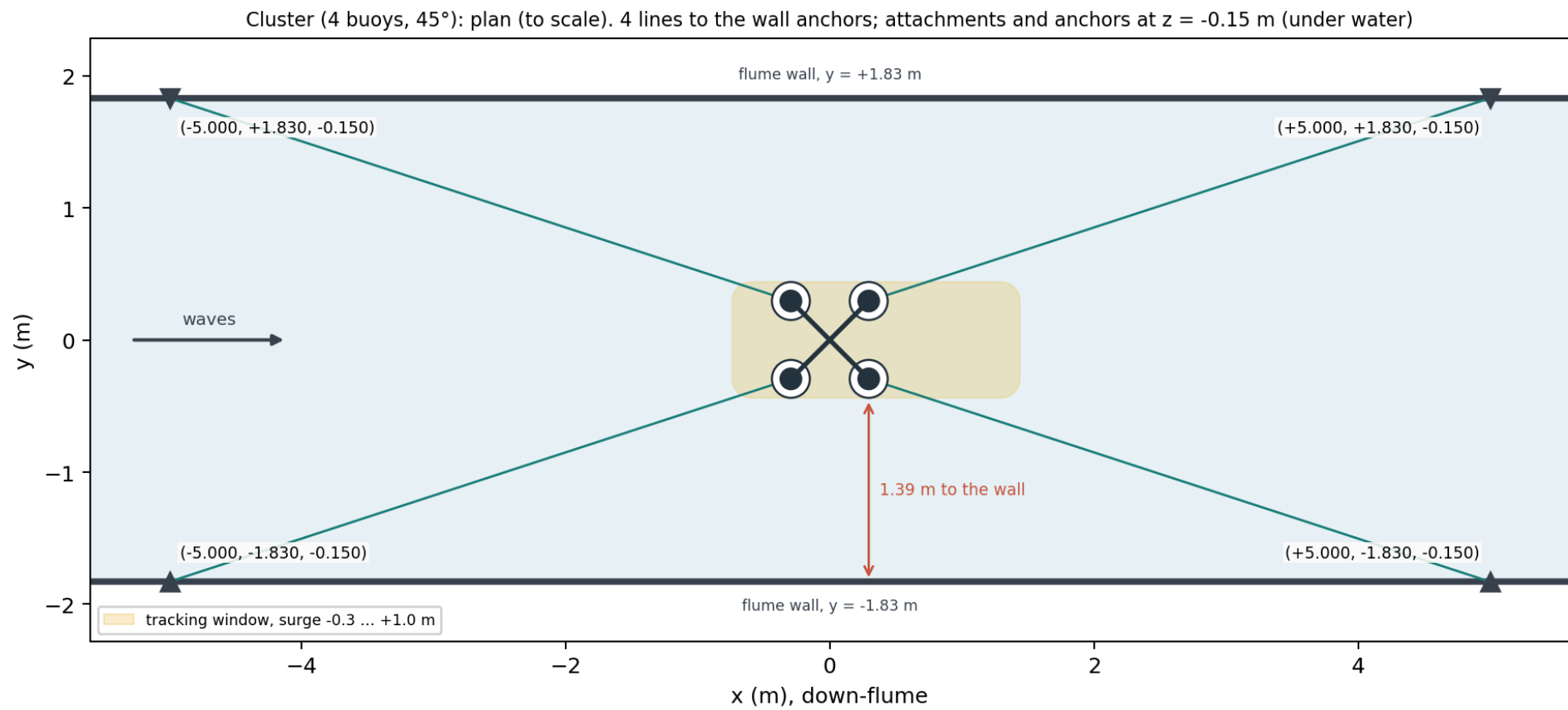
One cord set (response and extreme tests; its predicted $H = 0.5$ m offsets stay within ± 1 m). Peaks over every predicted run (\$5):

line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band -13 / +30 %)	pretension T_0 (at rest)	at-rest stretch	peak stretch	elongation capacity $\times 1.25$	max tension	working load $\times 3$
1 (buoy)	(-5.000, +1.830, -0.500)	(-0.188, +0.069, -0.501)	3.930 m	2.01 N/m (1.75–2.61)	2.40 N (2.40)	1195 mm	1.76 m	2.21 m (56 % of L0)	3.54 N	10.6 N
2 (buoy)	(-5.000, -1.830, -0.500)	(-0.188, -0.069, -0.501)	3.930 m	2.01 N/m (1.75–2.61)	2.40 N (2.40)	1195 mm	1.76 m	2.21 m (56 % of L0)	3.54 N	10.6 N
3 (buoy)	(+5.000, +1.830, -0.500)	(+0.188, +0.069, -0.501)	3.930 m	2.01 N/m (1.75–2.61)	2.40 N (2.40)	1195 mm	1.27 m	1.58 m (40 % of L0)	2.54 N	7.6 N
4 (buoy)	(+5.000, -1.830, -0.500)	(+0.188, -0.069, -0.501)	3.930 m	2.01 N/m (1.75–2.61)	2.40 N (2.40)	1195 mm	1.27 m	1.58 m (40 % of L0)	2.54 N	7.6 N

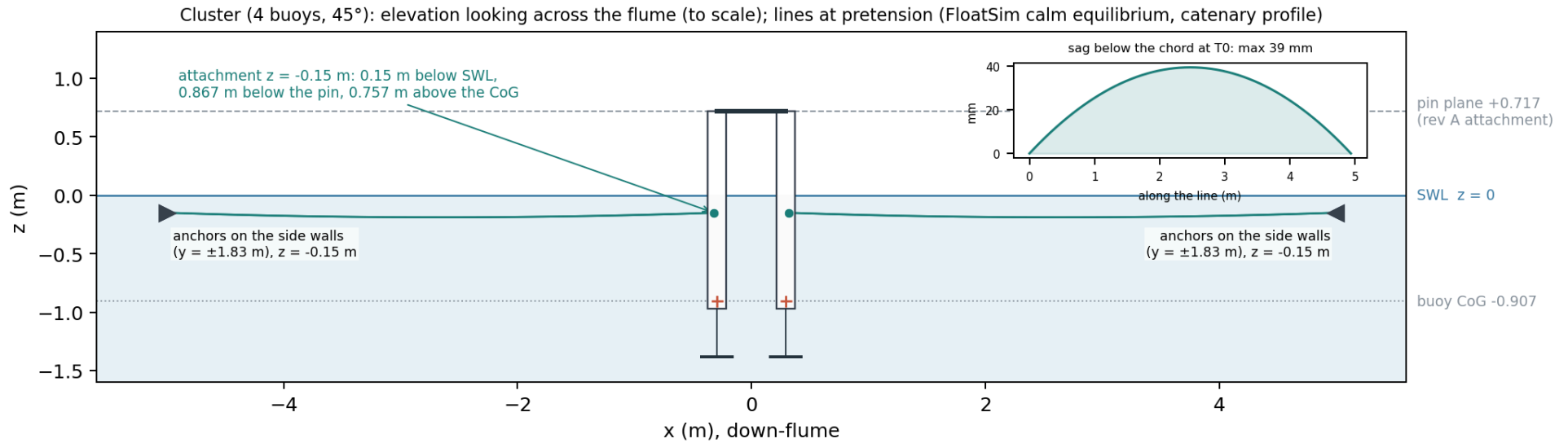
Anchors (working-load limit = $3 \times$ the larger design load of the cord sets):

anchor (x, y, z) m	lines	design load, the cord set	anchor working-load limit $\times 3$ (max of the sets)
(-5.000, +1.830, -0.500)	1	3.54 N	10.6 N
(-5.000, -1.830, -0.500)	2	3.54 N	10.6 N
(+5.000, +1.830, -0.500)	3	2.54 N	7.6 N
(+5.000, -1.830, -0.500)	4	2.54 N	7.6 N

2b. Cluster (4 buoys at 45°)



Cluster (4 buoys, 45°): plan, to scale (mooring_figures.py, from spec_statics.json)



Cluster (4 buoys, 45°): elevation looking across the flume, to scale

Operational cord set (response tests, $H \leq 0.12$ m). Peaks over its moored operational runs (§5):

line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band -13 / +30 %)	pretension T0 (at rest)	at-rest stretch	peak stretch	elongation capacity $\times 1.25$	max tension	working load $\times 3$
1 (buoy2)	(-5.000, +1.830, -0.150)	(-0.322, +0.303, -0.150)	4.589 m	4.15 N/m (3.61–5.40)	1.49 N (1.44)	346 mm	0.50 m	0.63 m (14 % of L0)	2.09 N	6.3 N
2 (buoy3)	(-5.000, -1.830, -0.150)	(-0.322, -0.303, -0.150)	4.589 m	4.15 N/m (3.61–5.40)	1.49 N (1.44)	346 mm	0.50 m	0.63 m (14 % of L0)	2.09 N	6.3 N
3 (buoy1)	(+5.000, +1.830, -0.150)	(+0.322, +0.303, -0.150)	4.589 m	4.15 N/m (3.61–5.40)	1.49 N (1.44)	346 mm	0.50 m	0.62 m (14 % of L0)	2.06 N	6.2 N
4 (buoy4)	(+5.000, -1.830, -0.150)	(+0.322, -0.303, -0.150)	4.589 m	4.15 N/m (3.61–5.40)	1.49 N (1.44)	346 mm	0.50 m	0.62 m (14 % of L0)	2.06 N	6.2 N

Extreme cord set, k $\times 5$ (load / survival tests, $H = 0.2\text{--}0.5$ m). Same anchors, attachments and nominal T0. Its stiffness is a MINIMUM: a softer cord puts $H = 0.5$ m past +1 m, so the band is $-0 / +30$ %. Peaks over its $H = 0.5$ m runs (§5):

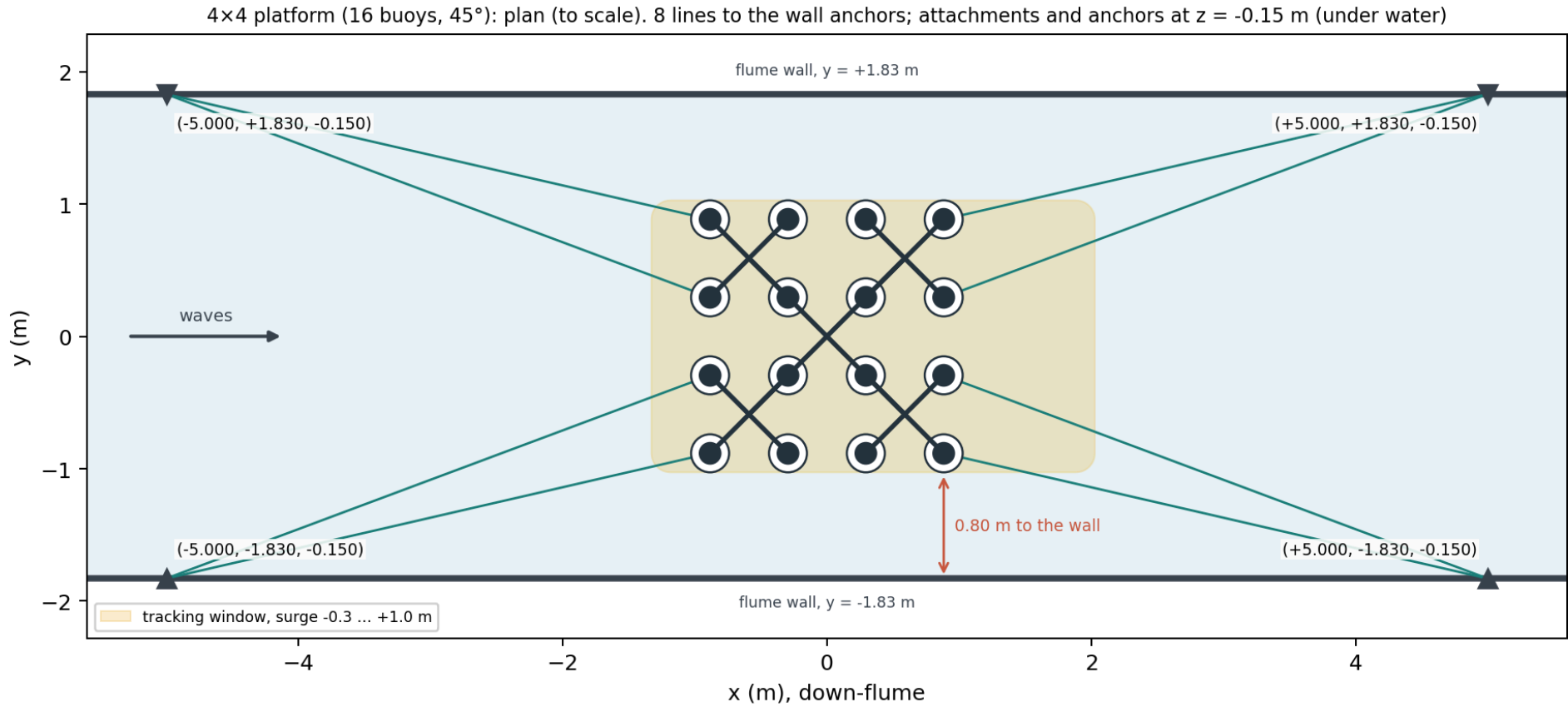
line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band +0 / +30 %)	pretension T0 (at rest)	at-rest stretch	peak stretch	elongation capacity $\times 1.25$	max tension	working load $\times 3$
1 (buoy2)	(-5.000, +1.830, -0.150)	(-0.322, +0.303, -0.150)	4.866 m	20.75 N/m (20.75–26.98)	1.73 N (1.44)	69 mm	1.07 m	1.33 m (27 % of L0)	22.12 N	66.4 N
2 (buoy3)	(-5.000, -1.830, -0.150)	(-0.322, -0.303, -0.150)	4.866 m	20.75 N/m (20.75–26.98)	1.73 N (1.44)	69 mm	1.07 m	1.33 m (27 % of L0)	22.12 N	66.4 N
3 (buoy1)	(+5.000, +1.830, -0.150)	(+0.322, +0.303, -0.150)	4.866 m	20.75 N/m (20.75–26.98)	1.73 N (1.44)	69 mm	0.00 m	0.01 m (0 % of L0)	0.10 N	0.3 N
4 (buoy4)	(+5.000, -1.830, -0.150)	(+0.322, -0.303, -0.150)	4.866 m	20.75 N/m (20.75–26.98)	1.73 N (1.44)	69 mm	0.00 m	0.01 m (0 % of L0)	0.10 N	0.3 N

Anchors (working-load limit = $3 \times$ the larger design load of the cord sets):

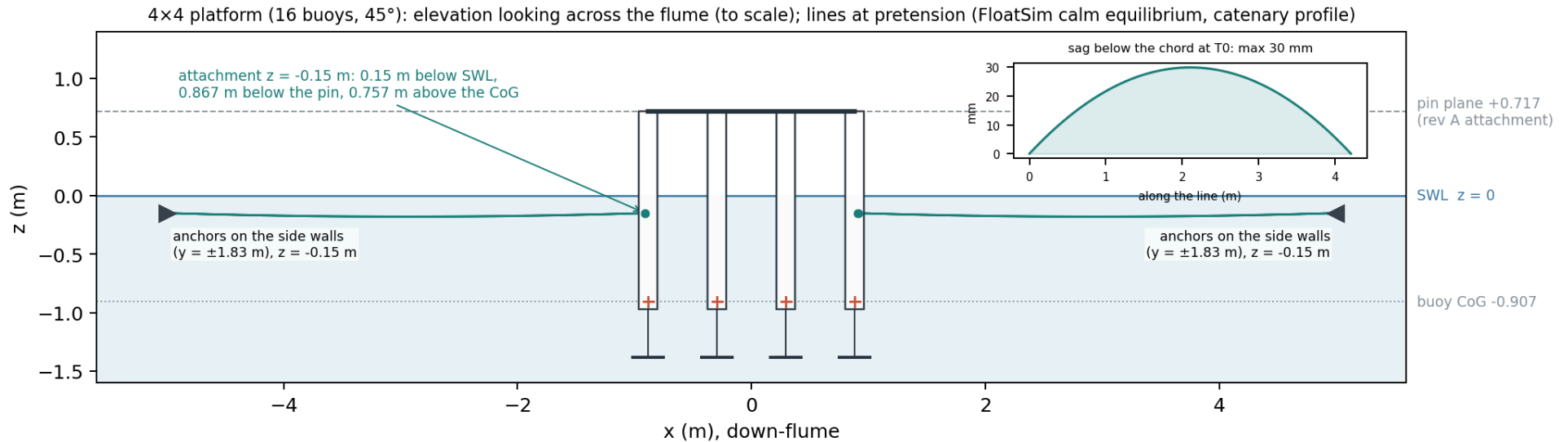
anchor (x, y, z) m	lines	design load, operational	design load, extreme	anchor working-load limit $\times 3$ (max of the sets)
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(-5.000, +1.830, -0.150)	1	2.09 N	22.12 N	66.4 N
(-5.000, -1.830, -0.150)	2	2.09 N	22.12 N	66.4 N
(+5.000, +1.830, -0.150)	3	2.06 N	0.10 N	6.2 N
(+5.000, -1.830, -0.150)	4	2.06 N	0.10 N	6.2 N

2c. 4×4 platform (16 buoys at 45°)



4×4 platform (16 buoys, 45°): plan, to scale (mooring_figures.py, from spec_statics.json)



4×4 platform (16 buoys, 45°): elevation looking across the flume, to scale

Operational cord set (response tests, $H \leq 0.12$ m). Peaks over its moored operational runs (§5):

line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band -13 / +30 %)	pretension T0 (at rest)	at-rest stretch	peak stretch	elongation capacity $\times 1.25$	max tension	working load $\times 3$
1 (buoy6)	(-5.000, +1.830, -0.150)	(-0.911, +0.890, -0.150)	4.038 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.34 m	0.42 m (10 % of L0)	2.81 N	8.4 N
2 (buoy7)	(-5.000, +1.830, -0.150)	(-0.910, +0.304, -0.150)	4.208 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.33 m	0.41 m (10 % of L0)	2.76 N	8.3 N
3 (buoy10)	(-5.000, -1.830, -0.150)	(-0.910, -0.304, -0.150)	4.208 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.33 m	0.41 m (10 % of L0)	2.76 N	8.3 N
4 (buoy11)	(-5.000, -1.830, -0.150)	(-0.911, -0.890, -0.150)	4.038 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.34 m	0.42 m (10 % of L0)	2.81 N	8.4 N
5 (buoy1)	(+5.000, +1.830, -0.150)	(+0.911, +0.890, -0.150)	4.038 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.31 m	0.39 m (10 % of L0)	2.60 N	7.8 N
6 (buoy4)	(+5.000, +1.830, -0.150)	(+0.910, +0.304, -0.150)	4.208 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.31 m	0.38 m (9 % of L0)	2.55 N	7.7 N
7 (buoy13)	(+5.000, -1.830, -0.150)	(+0.910, -0.304, -0.150)	4.208 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.31 m	0.38 m (9 % of L0)	2.55 N	7.7 N
8 (buoy16)	(+5.000, -1.830, -0.150)	(+0.911, -0.890, -0.150)	4.038 m	8.32 N/m (7.24–10.81)	1.54 N (1.42)	171 mm	0.31 m	0.39 m (10 % of L0)	2.60 N	7.8 N

Extreme cord set, k $\times 6$ (load / survival tests, $H = 0.2$ – 0.5 m). Same anchors, attachments and nominal T0. Its stiffness is a MINIMUM: a softer cord puts $H = 0.5$ m past +1 m, so the band is -0 / +30 %. Peaks over its $H = 0.5$ m runs (§5):

line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band +0 / +30 %)	pretension T0 (at rest)	at-rest stretch	peak stretch	elongation capacity $\times 1.25$	max tension	working load $\times 3$
1 (buoy6)	(-5.000, +1.830, -0.150)	(-0.911, +0.890, -0.150)	4.181 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.91 m	1.13 m (27 % of L0)	45.27 N	135.8 N
2 (buoy7)	(-5.000, +1.830, -0.150)	(-0.910, +0.304, -0.150)	4.350 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.88 m	1.10 m (25 % of L0)	43.87 N	131.6 N
3 (buoy10)	(-5.000, -1.830, -0.150)	(-0.910, -0.304, -0.150)	4.350 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.88 m	1.10 m (25 % of L0)	43.87 N	131.6 N

4 (buoy11)	(-5.000, -1.830, -0.150)	(-0.911, -0.890, -0.150)	4.181 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.91 m	1.13 m (27 % of L0)	45.27 N	135.8 N
5 (buoy1)	(+5.000, +1.830, -0.150)	(+0.911, +0.890, -0.150)	4.181 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.00 m	0.00 m (0 % of L0)	0.07 N	0.2 N
6 (buoy4)	(+5.000, +1.830, -0.150)	(+0.910, +0.304, -0.150)	4.350 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.00 m	0.00 m (0 % of L0)	0.08 N	0.2 N
7 (buoy13)	(+5.000, -1.830, -0.150)	(+0.910, -0.304, -0.150)	4.350 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.00 m	0.00 m (0 % of L0)	0.08 N	0.2 N
8 (buoy16)	(+5.000, -1.830, -0.150)	(+0.911, -0.890, -0.150)	4.181 m	49.91 N/m (49.91–64.88)	2.14 N (1.42)	28 mm	0.00 m	0.00 m (0 % of L0)	0.07 N	0.2 N

Anchors (working-load limit = 3 × the larger design load of the cord sets):

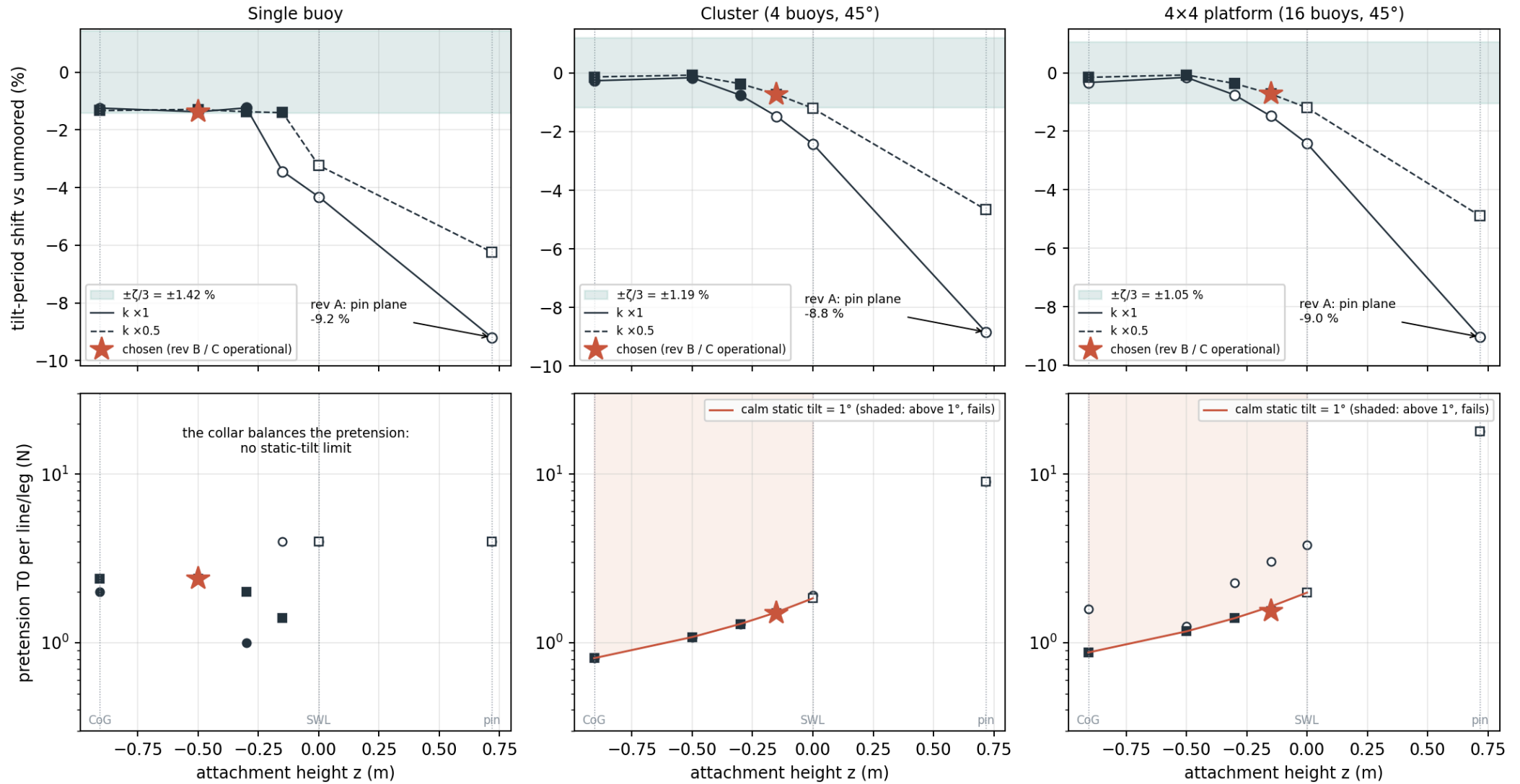
anchor (x, y, z) m	lines	design load, operational	design load, extreme	anchor working-load limit ×3 (max of the sets)
(-5.000, +1.830, -0.150)	1, 2	5.57 N	89.14 N	267.4 N
(-5.000, -1.830, -0.150)	3, 4	5.57 N	89.14 N	267.4 N
(+5.000, +1.830, -0.150)	5, 6	5.15 N	0.14 N	15.5 N
(+5.000, -1.830, -0.150)	7, 8	5.15 N	0.14 N	15.5 N

3. Design: the attachment height (rev B) and the extreme cord set (rev C)

3a. Attachment height and the operational set

- Rev A attached at the pin plane (+0.717 m) to remove calm-water tilt. At the design pretension and stiffness its tilt period shifts **buoy -9.2 % against ± 1.42 %, cluster -8.8 % against ± 1.19 %, platform -9.0 % against ± 1.05 %**, several times what the resonance tolerates.
- The two effects act about different points:
- The static pretension moment acts about the PIN.
- The dynamic coupling ($\approx k \cdot h^2$) acts about the tilt mode's rotation centre, near the CoG.
- **The rule** (select, do not ask). HARD criteria:
- tilt-period shift vs the unmoored article $\leq \zeta/3$ (ζ at $H = 0.04$ m);
- heave shift ≤ 1 %;
- calm static tilt $\leq 1^\circ$ (now confirmed);
- surge ≥ 14 s;
- no slack in the operational band with the drift sum.
- Selection: the most pretension, then the attachment closest to the waterline, then the stiffest surge.

Design rationale (attachment_sweep.py): each point is the sweep table's row, i.e. the most pretension allowed at that height (filled = passes every hard criterion)



Design rationale: tilt-period shift and pretension against attachment height (the sweep table's rows)

The sweep. Each row is the most pretension that passes at that height and k . Where none passes, it is the most that keeps the static tilt and the operational band, and the row shows what fails.

The static tilt is the estimate, except in the chosen rows (FloatSim's settle):

article	attachment z	k	T0 per line/leg	tilt shift (limit)	heave shift	calm static tilt	surge	op. T_min/T0	verdict
buoy	pin +0.717 m	$\times 1$	4.00 N	-9.19 % (1.42)	-0.83 %	0.00°	16.4 s	0.80	fails: tilt -9.19 %

buoy	pin +0.717 m	×0.5	4.00 N	-6.23 % (1.42)	-0.83 %	0.00°	22.6 s	0.90	fails: tilt -6.23 %
buoy	+0.00 m	×1	4.00 N	-4.32 % (1.42)	-0.83 %	0.00°	16.4 s	0.90	fails: tilt -4.32 %
buoy	+0.00 m	×0.5	4.00 N	-3.23 % (1.42)	-0.83 %	0.00°	22.6 s	0.95	fails: tilt -3.23 %
buoy	-0.15 m	×1	4.00 N	-3.43 % (1.42)	-0.83 %	0.00°	16.4 s	0.92	fails: tilt -3.43 %
buoy	-0.15 m	×0.5	1.40 N	-1.40 % (1.42)	-0.29 %	0.00°	23.4 s	0.88	pass
buoy	-0.30 m	×1	1.00 N	-1.23 % (1.42)	-0.21 %	0.00°	16.8 s	0.75	pass
buoy	-0.30 m	×0.5	2.00 N	-1.37 % (1.42)	-0.42 %	0.00°	23.2 s	0.94	pass
buoy	-0.50 m	×1	2.40 N	-1.37 % (1.42)	-0.50 %	0.00°	16.6 s	0.94	CHOSEN
buoy	-0.50 m	×0.5	2.40 N	-1.29 % (1.42)	-0.50 %	0.00°	23.1 s	0.96	pass
buoy	-0.91 (CoG) m	×1	2.00 N	-1.24 % (1.42)	-0.42 %	0.00°	16.6 s	0.94	pass
buoy	-0.91 (CoG) m	×0.5	2.40 N	-1.33 % (1.42)	-0.50 %	0.00°	23.1 s	0.96	pass
cluster	pin +0.717 m	×1	9.01 N	-8.83 % (1.19)	-0.49 %	0.00°	16.2 s	0.63	fails: tilt -8.83 %
cluster	pin +0.717 m	×0.5	9.01 N	-4.66 % (1.19)	-0.49 %	0.00°	22.7 s	0.82	fails: tilt -4.66 %
cluster	+0.00 m	×1	1.92 N	-2.42 % (1.19)	-0.10 %	1.05°	16.4 s	0.15	fails: tilt -2.42 %, static 1.05°, op slack 0.1498 (even the least T0 that keeps the operational band taut)
cluster	+0.00 m	×0.5	1.83 N	-1.21 % (1.19)	-0.10 %	1.00°	23.1 s	0.55	fails: tilt -1.21 %
cluster	-0.15 m	×1	1.52 N	-1.48 % (1.19)	-0.08 %	1.00°	16.4 s	0.15	fails: tilt -1.48 %
cluster	-0.15 m	×0.5	1.49 N	-0.73 % (1.19)	-0.08 %	1.00°	23.1 s	0.56	CHOSEN
cluster	-0.30 m	×1	1.29 N	-0.76 % (1.19)	-0.07 %	1.00°	16.5 s	0.26	pass
cluster	-0.30 m	×0.5	1.29 N	-0.37 % (1.19)	-0.07 %	1.00°	23.2 s	0.63	pass
cluster	-0.50 m	×1	1.08 N	-0.16 % (1.19)	-0.06 %	1.00°	16.5 s	0.52	pass
cluster	-0.50 m	×0.5	1.08 N	-0.07 % (1.19)	-0.06 %	1.00°	23.2 s	0.66	pass
cluster	-0.91 (CoG) m	×1	0.81 N	-0.26 % (1.19)	-0.05 %	1.00°	16.8 s	0.24	pass
cluster	-0.91 (CoG) m	×0.5	0.81 N	-0.13 % (1.19)	-0.04 %	1.00°	23.4 s	0.58	pass
platform	pin +0.717 m	×1	18.03 N	-9.04 % (1.05)	-0.10 %	0.00°	16.1 s	0.63	fails: tilt -9.04 %
platform	pin +0.717 m	×0.5	18.03 N	-4.89 % (1.05)	-0.10 %	0.00°	22.5 s	0.82	fails: tilt -4.89 %
platform	+0.00 m	×1	3.79 N	-2.42 % (1.05)	-0.05 %	1.91°	16.2 s	0.15	fails: tilt -2.42 %, static 1.91° (even the least T0 that keeps the operational band taut)
platform	+0.00 m	×0.5	1.98 N	-1.19 % (1.05)	-0.03 %	1.00°	23.0 s	0.18	fails: tilt -1.19 %
platform	-0.15 m	×1	3.02 N	-1.48 % (1.05)	-0.04 %	1.84°	16.2 s	0.15	fails: tilt -1.48 %, static 1.84° (even the least T0 that keeps the operational band taut)
platform	-0.15 m	×0.5	1.54 N	-0.72 % (1.05)	-0.03 %	1.00°	23.0 s	0.18	CHOSEN
platform	-0.30 m	×1	2.26 N	-0.76 % (1.05)	-0.03 %	1.61°	16.3 s	0.15	fails: static 1.61° (even the least T0 that keeps the operational band taut)
platform	-0.30 m	×0.5	1.40 N	-0.36 % (1.05)	-0.02 %	1.00°	23.0 s	0.34	pass
platform	-0.50 m	×1	1.26 N	-0.16 % (1.05)	-0.02 %	1.08°	16.4 s	0.15	fails: static 1.08° (even the least T0 that keeps the operational band taut)
platform	-0.50 m	×0.5	1.17 N	-0.07 % (1.05)	-0.02 %	1.00°	23.1 s	0.50	pass
platform	-0.91 (CoG) m	×1	1.59 N	-0.33 % (1.05)	-0.03 %	1.82°	16.3 s	0.15	fails: static 1.82°, op slack 0.1485 (even the least T0 that keeps the operational band taut)
platform	-0.91 (CoG) m	×0.5	0.88 N	-0.15 % (1.05)	-0.02 %	1.00°	23.3 s	0.18	pass

- **Residual interference:**

- The operational set still shifts the tilt resonance, within $\zeta/3$ and modelled by FloatSim.
- That is acceptable for validating FloatSim.

- It **must be declared** in any claim about the free (unmoored) response.
- ⚠ **The buoy's yaw vs tilt conflict:**
- The collar's pretension stiffens pitch and yaw together ($\propto T_0 \cdot r$), so every buoy design that passes $\zeta/3$ has a yaw period ≥ 1.12 s.
- FloatSim finds part of the buoy's extreme band unstable in yaw (§6).
- A pitch-neutral collar (e.g. a cross-flume bar) is a follow-up.

3b. The extreme cord set

Why: the operational set at H = 0.5 m (rev B's runs):

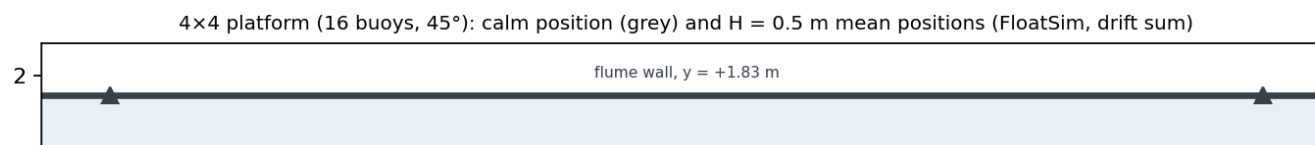
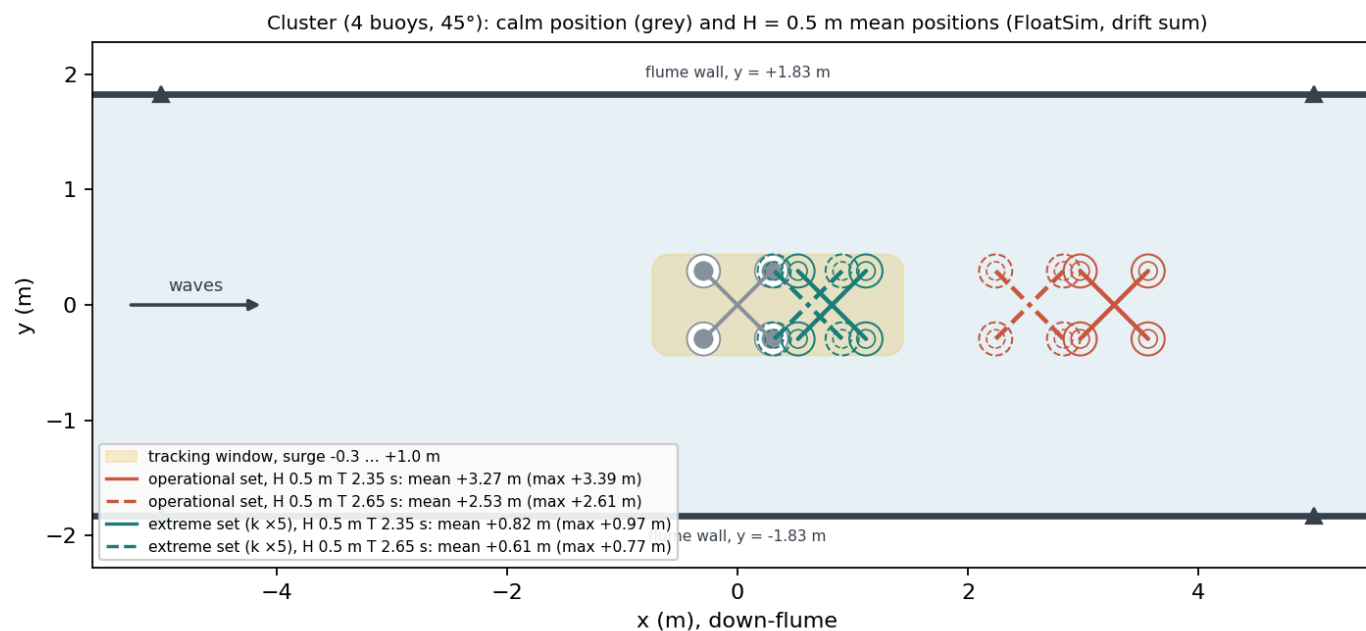
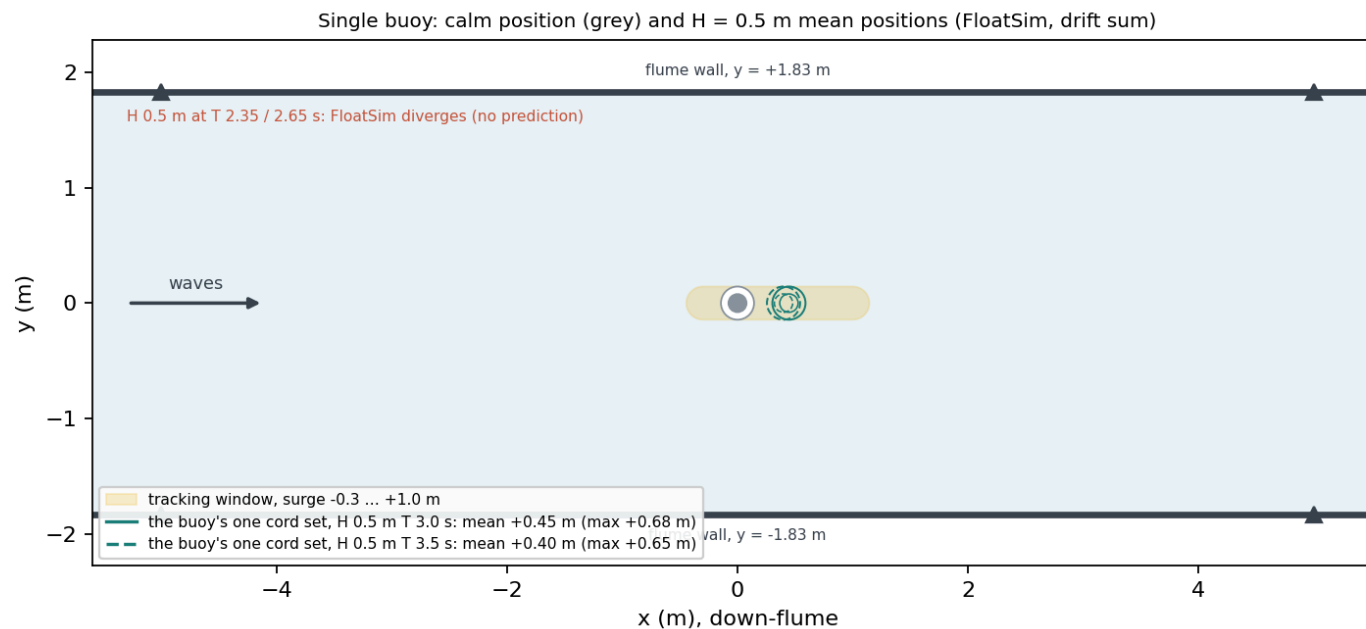
article	set	case (drift sum)	min / max line tension	slack lines: peak re-tension	mean offset (range)	max tilt	max yaw	line above the local surface	validity
cluster	operational (rev B)	H 0.5 m, T 2.35 s	0.04 / 15.23 N	stay slack (≤ 0.05 N, their hanging weight): no re-tension in the wave train	3.27 m (3.15–3.39) > +1 m	15.3°	0.70°	breaks the surface by 0.19 m	indicative (tilt > 5.7°)
cluster	operational (rev B)	H 0.5 m, T 2.65 s	0.05 / 12.01 N	stay slack (≤ 0.05 N, their hanging weight): no re-tension in the wave train	2.53 m (2.46–2.61) > +1 m	12.9°	0.47°	breaks the surface by 0.18 m	indicative (tilt > 5.7°)
platform	operational (rev B)	H 0.5 m, T 2.35 s	0.04 / 29.43 N	stay slack (≤ 0.04 N, their hanging weight): no re-tension in the wave train	3.41 m (3.30–3.53) > +1 m	22.7°	1.78°	breaks the surface by 0.18 m	indicative (tilt > 5.7°)
platform	operational (rev B)	H 0.5 m, T 2.65 s	0.04 / 24.70 N	stay slack (≤ 0.05 N, their hanging weight): no re-tension in the wave train	2.85 m (2.78–2.93) > +1 m	19.4°	1.38°	breaks the surface by 0.17 m	indicative (tilt > 5.7°)

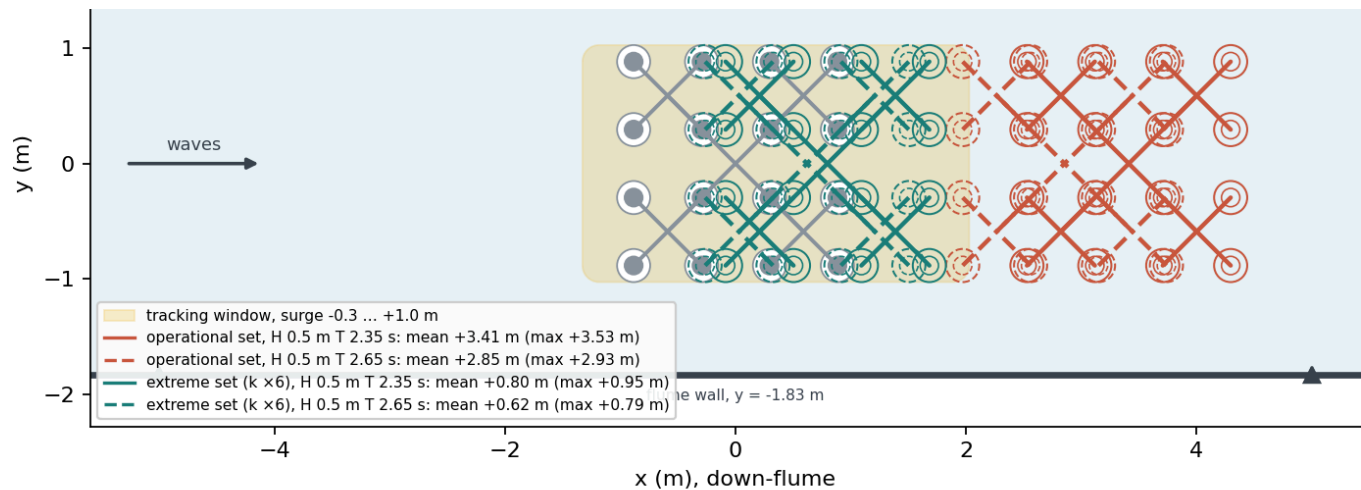
Selection:

- The rule: the smallest cord stiffness, as a multiple of the operational k, that keeps the article inside the tracking window at H = 0.5 m. It is checked at T = 2.35 s (the largest admissible drift) and 2.65 s (the tilt/heave resonance), with the conservative drift sum applied in-run (FloatSim).
- Two readings of "inside the window":
- The confirmed criterion 4 is mean offset + dynamic amplitude ≤ 1.0 m, i.e. the run's maximum surge. **The spec uses it.**
- The mean offset alone (Xabier's wording) would allow $\times 4.5$ (cluster) and $\times 5$ (platform).
- Chosen: **cluster $\times 5$, platform $\times 6$.**

article	k (× operational)	T = 2.35 s: mean / max surge	T = 2.65 s: mean / max surge	peak line tension	slack lines: peak re-tension	verdict (max surge ≤ 1.0 m, both periods)
cluster	×3.5 (14.53 N/m)	1.18 / 1.30 m	0.87 / 1.02 m	20.02 N	0.07 N	fails
cluster	×4 (16.60 N/m)	1.04 / 1.17 m	0.77 / 0.92 m	20.80 N	0.08 N	fails
cluster	×4.5 (18.68 N/m)	0.93 / 1.07 m	0.69 / 0.84 m	21.48 N	0.09 N	fails (mean alone passes)
cluster	×5 (20.75 N/m)	0.83 / 0.98 m	0.62 / 0.78 m	22.12 N	0.09 N	CHOSEN
cluster	×5.5 (22.83 N/m)	0.76 / 0.91 m	0.57 / 0.73 m	22.76 N	0.10 N	pass
cluster	×6 (24.90 N/m)	0.69 / 0.85 m	0.53 / 0.68 m	23.44 N	0.12 N	pass
cluster	×5 FINAL (at-rest-matched T0)	0.82 / 0.97 m	0.61 / 0.77 m	22.12 N	0.10 N	PASS: the spec
platform	×3.5 (29.11 N/m)	1.30 / 1.41 m	1.00 / 1.15 m	38.60 N	0.05 N	fails
platform	×4 (33.27 N/m)	1.16 / 1.29 m	0.89 / 1.05 m	40.09 N	0.06 N	fails
platform	×4.5 (37.43 N/m)	1.05 / 1.18 m	0.80 / 0.97 m	41.44 N	0.06 N	fails
platform	×5 (41.59 N/m)	0.95 / 1.09 m	0.74 / 0.90 m	42.72 N	0.07 N	fails (mean alone passes)
platform	×5.5 (45.75 N/m)	0.88 / 1.02 m	0.68 / 0.85 m	43.99 N	0.07 N	fails (mean alone passes)
platform	×6 (49.91 N/m)	0.81 / 0.96 m	0.63 / 0.81 m	45.28 N	0.07 N	CHOSEN

platform	×6 FINAL (at-rest-matched T0)	0.80 / 0.95 m	0.62 / 0.79 m	45.27 N	0.08 N	PASS: the spec
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$H = 0.5$ m mean positions of both cord sets against the tracking window (FloatSim, drift sum)

- **Declared, not criteria:** these tests measure loads and survival, and the tank and the model see the same mooring. The extreme set's modal periods against the unmoored article: cluster tilt -3.77 %, heave -0.09 %, surge 10.4 s, platform tilt -4.65 %, heave -0.03 %, surge 9.4 s.
- Δ The surge period is below the confirmed criterion 2 (≥ 14 s), so the wave-frequency surge is amplified by $1/(1 - (T/T_{\text{surge}})^2)$: cluster +13 % (T_{surge} 10.5 s) / platform +15 % (T_{surge} 9.6 s) at $T = 3.5$ s.
- The regular-wave matrix has no difference-frequency forcing.
- Criteria 2 and 4 cannot both hold for this set at $H = 0.5$ m: ≥ 14 s caps the stiffness at about $\times 2.8$, which leaves ≥ 1.5 m of offset (extrapolated from the k scan).
- **Pretension: the same AT-REST tension, not the same nominal T0.**
- The code's nominal T0 is the tension at the design (untitled) chord. The calm 1° tilt moves each attachment $\delta \approx 15$ mm towards its anchor, so the at-rest tension is $T_0 - k \cdot \delta$, and a stiffer cord loses more.
- The extreme set is therefore tensioned to the operational set's at-rest tension. Its line forces at the calm geometry are then identical, so **the operational settle is its calm equilibrium: the same 1.00° / 1.00° calm tilt, no new settle**. FloatSim's joint-projected static residual at the operational settle: cluster 0.0077 N with the operational lines, 0.0073 N with the extreme lines (nominal T0 instead: 0.13 N); platform 0.0024 N with the operational lines, 0.0022 N with the extreme lines (nominal T0 instead: 0.24 N) (extreme_set_residuals.json).
- Its L0 and nominal T0 in the tables implement this.
- **Raising T0** (optional): Not run, because there is nothing to cut. In the final runs the down-flume lines go slack and **stay slack through the wave train** (their peak is their hanging weight). Their pre-stretch is far below the mean offset, and it stays so even at the 3° calm-tilt cap ($T_0 \approx 3 \times$ the at-rest tension: pre-stretch cluster 208 mm against a mean offset ≥ 0.61 m; platform 85 mm against a mean offset ≥ 0.62 m). Raising T0 would only add to the up-flume peak.

4. Hardware

- **Elastic cord, usable submerged: two sets for the cluster and platform, one for the buoy.**
- It must hold its secant stiffness over the working stretch (at-rest to peak):
- the operational set within $-13 / +30$ % of the target;
- the extreme set **at or above** its target, up to $+30$ %: softer moves the $H = 0.5$ m cases out of the tracking window.
- Required elongation capacity: the tables ($\times 1.25$ on the FloatSim peak stretch). If the cord is shorter than L_0 (a stiff lead in series), its strain rises in proportion.
- Hysteresis is not modelled. It damps the slow modes and shifts the mean position by half the loading/unloading gap, which the pull test measures.
- **Pulley and counterweight (constant tension): rejected** (rev A §3). All its restoring is geometric, so it is far too soft in surge.
- **Buoy collar: a SLENDER spreader.** The collar is under water, an appendage the model does not include:
- It has 4 arms of $r = 0.2$ m, 0.8 m in total. The added-mass budget is **0.4 kg**, so a round section is at most **25 mm** in diameter (added mass $\approx \rho \pi D^2/4$ per metre in heave).
- A 10 mm rod adds 0.06 kg and carries the peak line tension (3.54 N at the arm tip) at ≈ 7 MPa of bending stress at the root, fine in aluminium or steel. A tensioned wire spreader is also acceptable.
- **A disk is NOT acceptable:** a solid disk of $r = 0.2$ m would add ≈ 21 kg of heave added mass (the heave plate's own is ≈ 7.9 kg).

5. Installation checks (each cord set, installed, before any waves)

- **Calm static tilt: the pretension check.**
- On the cluster and platform the calm tilt IS the pretension gauge: FloatSim gives 1.00° / 1.00° (operational) and 1.00° / 1.00° (extreme) at the design T0 (cluster / platform).
- It is linear in the at-rest tension: $\approx 0.70^\circ/\text{N}$ / $0.70^\circ/\text{N}$.
- The extreme set's pre-stretch is only 69 mm / 28 mm, so a length error of a few mm is a large T0 error. **Set the extreme set's T0 by tension or by the calm tilt, not by length.**
- The buoy's collar balances the pretension (calm tilt 0): measure its line tension directly.
- **Static pull and decays:**
- the static pull is a horizontal pull through the article centre at the attachment depth, force against offset;
- the free decays start from ~ 0.3 m surge and ~ 0.1 m sway;
- both are compared with the FloatSim predictions below.
- The buoy's yaw decay is not predictable in FloatSim.

article	cord set	surge period	sway period	yaw period	surge pull K / F at 0.25, 0.5 m	sway pull K / F at 0.1 m	yaw pull K / M at 5°	calm static tilt (FloatSim settle)
buoy	one set	16.6 s	26.4 s	1.12 s	7.31 N/m / 1.83, 3.65 N	2.60 N/m / 0.260 N	2.00 N·m/rad / 0.174 N·m	0.00°
cluster	operational	23.1 s	51.6 s	24.78 s	15.05 N/m / 3.72, 6.36 N	2.64 N/m / 0.264 N	3.06 N·m/rad / 0.268 N·m	1.00°
cluster	extreme	10.5 s	28.2 s	18.75 s	73.17 N/m / 11.96, 21.61 N	8.83 N/m / 0.877 N	5.35 N·m/rad / 0.471 N·m	1.00°
platform	operational	23.0 s	59.0 s	33.40 s	60.63 N/m / 12.65, 20.66 N	8.11 N/m / 0.812 N	28.96 N·m/rad / 2.547 N·m	1.00°
platform	extreme	9.6 s	28.3 s	18.08 s	349.33 N/m / 50.99, 97.63 N	35.30 N/m / 3.189 N	98.86 N·m/rad / 7.512 N·m	1.00°

The surge and sway periods are the round-1 FloatSim decays scaled by $\sqrt{(K_{\text{decay}} / K)}$ (same effective mass).

6. Predicted envelope

Operational set, operational band (cluster and platform; FloatSim, the moored design, the drift sum):

article	case (drift sum)	min / max line tension	T _{min} / T ₀ (≥ 0.15)	mean offset (range)	max tilt	max yaw	validity
cluster	H 0.12 m, T 1.4 s	1.04 / 1.83 N	0.73 PASS	0.07 m (0.05–0.09)	2.5°	7.9e-04°	LEVEL1 valid
cluster	H 0.12 m, T 2.84 s	0.81 / 2.09 N	0.56 PASS	-0.00 m (-0.04–0.04)	17.6°	2.1e-02°	indicative (tilt > 5.7°)
platform	H 0.12 m, T 1.4 s	0.87 / 1.99 N	0.61 PASS	0.06 m (0.05–0.07)	2.2°	3.2e-03°	LEVEL1 valid
platform	H 0.12 m, T 2.9 s	0.27 / 2.81 N	0.19 PASS	0.00 m (-0.05–0.06)	17.5°	3.6e-02°	indicative (tilt > 5.7°)

Extreme set, H = 0.5 m (cluster and platform; FloatSim, the drift sum):

article	set	case (drift sum)	min / max line tension	slack lines: peak re-tension	mean offset (range)	max tilt	max yaw	line above the local surface	validity
cluster	extreme, k ×5	H 0.5 m, T 2.35 s	0.06 / 22.12 N	stay slack (≤ 0.07 N, their hanging weight): no re-tension in the wave train	0.82 m (0.66–0.97)	21.6°	0.92°	breaks the surface by 0.18 m	indicative (tilt > 5.7°)
cluster	extreme, k ×5	H 0.5 m, T 2.65 s	0.06 / 20.28 N	stay slack (≤ 0.10 N, their hanging weight): no re-tension in the wave train	0.61 m (0.45–0.77)	30.5°	0.81°	breaks the surface by 0.17 m	indicative (tilt > 5.7°)
platform	extreme, k ×6	H 0.5 m, T 2.35 s	0.05 / 45.27 N	stay slack (≤ 0.06 N, their hanging weight): no re-tension in the wave train	0.80 m (0.64–0.95)	27.9°	2.55°	breaks the surface by 0.17 m	indicative (tilt > 5.7°)
platform	extreme, k ×6	H 0.5 m, T 2.65 s	0.05 / 44.69 N	stay slack (≤ 0.08 N, their hanging weight): no re-tension in the wave train	0.62 m (0.44–0.79)	32.9°	2.32°	breaks the surface by 0.16 m	indicative (tilt > 5.7°)

Single buoy (one cord set; FloatSim, the drift sum, dt = 0.0025 s, 210 s).

- *stable*: yaw at round-off.
- *yaw grows*: the model grows yaw, still < 1° at the end of the run.
- *diverges*: no prediction.

H	T	FloatSim	max tilt	max yaw	T _{min} / T ₀	max tension	slack lines: peak re-tension	surge range
0.04 m	2.75 s	stable	9.4°	3.6e-11°	0.97	2.46 N	none	-0.04–0.04 m
0.12 m	2.45 s	stable	9.5°	2.9e-11°	0.94	2.55 N	none	-0.01–0.01 m
0.12 m	2.55 s	stable	12.7°	1.1e-09°	0.93	2.57 N	none	-0.01–0.02 m
0.12 m	2.65 s	stable	15.8°	1.5e-10°	0.93	2.57 N	none	-0.04–0.05 m
0.12 m	2.75 s	stable	16.8°	1.1e-09°	0.95	2.53 N	none	-0.07–0.07 m
0.2 m	2.35 s	stable	11.9°	4.3e-11°	0.88	2.69 N	none	0.02–0.07 m
0.2 m	2.65 s	yaw grows	20.9°	8.2e-05°	0.88	2.71 N	none	-0.01–0.10 m
0.2 m	2.75 s	yaw grows	22.0°	2.0e-03°	0.91	2.62 N	none	-0.07–0.11 m
0.2 m	3.5 s	stable	5.2°	1.2e-11°	0.93	2.57 N	none	-0.08–0.13 m
0.35 m	2.0 s	yaw grows	10.9°	2.2e-05°	0.53	3.54 N	none	0.40–0.52 m
0.35 m	2.35 s	yaw grows	18.0°	1.3e-05°	0.60	3.38 N	none	0.29–0.39 m
0.35 m	2.65 s	diverges: no prediction	—	—	—	—	—	—

0.35 m	3.0 s	yaw grows	23.4°	2.0e-06°	0.85	2.76 N	none	-0.08–0.30 m
0.35 m	3.5 s	stable	9.4°	4.7e-11°	0.80	2.88 N	none	-0.05–0.31 m
0.5 m	2.35 s	diverges: no prediction	—	—	—	—	—	—
0.5 m	2.65 s	diverges: no prediction	—	—	—	—	—	—
0.5 m	3.0 s	yaw grows	29.5°	2.5e-06°	0.56	3.50 N	none	0.22–0.68 m
0.5 m	3.5 s	stable	14.0°	4.8e-11°	0.55	3.49 N	none	0.15–0.65 m

- **Operational band ($H \leq 0.12$ m): no slack, for every article.**
- **Small-angle validity:** FloatSim here is LEVEL1 (valid to 0.1 rad = 5.7°).
- Every prediction near the tilt resonance, line loads included, is **indicative**.
- All the extremes are beyond it. LEVEL2 decides whether the buoy's divergence is physical.
- **Snap loads are not modelled.** FloatSim's line is quasi-static (no line inertia). The "peak re-tension" is the quasi-static tension when a slack line comes taut again. A real snatch can exceed it.
- **Yaw of the cluster and platform in the extremes.**
- It is forced by the 1e-5 asymmetry of the coupled radiation kernel (DESIGN-BASIS §C6).
- Only the peak was kept, so whether it is steady or growing is not established.
- Phase D's proposed flag is 0.5°.

7. Test procedure

1. **Install the operational set.** Check each line's T0 through the calm tilt (cluster and platform) or the tension (buoy). Then run the static pull and the decays (§5).
2. **Response series ($H \leq 0.12$ m)** with the operational set.
3. **Swap the cords to the extreme set** at the same anchors, attachment points and at-rest tension. Set it by tension or calm tilt (1°), then repeat the §5 checks against the extreme-set rows.
4. **Extreme series in increasing H** ($0.2 \rightarrow 0.35 \rightarrow 0.5$ m) at each period, with live offset monitoring. Stop a case if the down-flume offset approaches the tracking limit (e.g. 0.9 m of the +1.0 m) or a line nears its working load. Slack of the down-flume lines is allowed.
5. **The buoy's extremes LAST** (it has one cord set), with a height ramp ($0.2 \rightarrow 0.35 \rightarrow 0.5$ m). Watch yaw and roll, and **stop on any growth of yaw**. Three cases ($H = 0.35$ m at 2.65 s, $H = 0.5$ m at 2.35 and 2.65 s) have no reliable FloatSim prediction.
6. Heading 0 (waves along the flume axis) for every article. Skip cases with $H/\lambda > 0.08$, and $H = 0.5$ m below $T = 2.35$ s.

8. Follow-ups before the test

Item	What it could change
The extreme set's surge period (below criterion 2's 14 s; declared here)	Xabier to accept the declaration. The alternative, $\leq \times 2.8$ stiffness, leaves the $H = 0.5$ m cases ≥ 1.5 m down-flume (extrapolated).
Return swing after the wave train	The down-flume lines stay slack during the train and re-tension when the drift stops and the article swings back. Energy bounds the swing to the mean offset, so their tension stays below the up-flume peak and the same working load covers it. Not simulated.
Drift mechanism	The sum used today adds FloatSim's own mean force to the recorded fixed-body splash-zone bound. A moving-body splash-zone model could lower the offsets and tensions, and so the extreme set's stiffness.
BEM regeneration at NT = 36 (flume databases' 12-sided waterline; ~1.7 h platform BEM)	All heave/tilt resonances ~0.04–0.08 s shorter. The $\zeta/3$ comparison holds (it is relative to the unmoored article).
Full Phase D , both cord sets	The criteria across the whole matrix. The extreme set is verified here only at $H = 0.5$ m, $T = 2.35$ and 2.65 s.
LEVEL2 rerun of the resonant subset	Resonant tilts, line loads and offsets, all indicative today. It also decides the buoy's divergence.
Buoy yaw fix (a pitch-neutral collar, e.g. a cross-flume bar)	Could restore the buoy's $H \geq 0.35$ m extremes near resonance without the tilt shift.
Snatch loads of the extreme set	FloatSim's quasi-static line gives the re-tension tension only. A dynamic line model, or a pull test with a load cell, bounds the snatch.
HWRL confirmations F1–F7	Any of them can force a redesign.